

PROJECT ANNOUNCEMENT

Fall 2026

Title: BATTERY THERMAL MANAGEMENT SYSTEM USING PCM IN ELECTRIC VEHICLES BY CFD ANALYSIS

Position Ref: TMS

Type: Group/Individual

Status: New/Ongoing

Seats: 1/2 student(s)

Project Description:

The battery cooling systems increase the driving range of electric vehicles. At the same time, it maintains Li-ion battery temperatures in the optimal working range (25 °C - 55 °C), assuring required efficiency of the cells and modules and preventing thermal runaway and related safety issues. In this project, typical prismatic lithium-ion cells will be simulated using numerical methods such as ANSYS or COMSOL. Air and liquid cooling systems will be simulated and compared for electric car driving conditions.

Work Description:

- Literature review on thermal management systems
- Active cooling systems for different battery types, especially Li-ion batteries
- Defining a proper geometry
- Specifying the mesh and boundary conditions
- Solving the problem in a proper platform, preferably ANSYS
- Post-processing of the results and concluding remarks

Required Qualifications and Skills:

- Basic knowledge of heat transfer
- Simulation experience in CFD or similar programs would be helpful
- Drawing capability (for creating the geometry)

Related UN Sustainable Development Goals:

•SDG 7 – Affordable and Clean Energy:

Difficulty: high () medium (X) low ()

Time commitment: high () medium (X) low ()

Estimated expense: 0 TL